EAS109

Examination - January Semester 2024

Thermo-Fluid Mechanics

Thursday, 18 April 2024 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR** (**4**) questions and comprises **SEVEN** (**7**) printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Full marks: 100)

Question 1

In a car testing facility (see Figure Q1), air is drawn from the ambient (Point 1) into a wind tunnel at a speed of 97 km/h. A manometer is used to measure the static pressure at the wind tunnel wall using water of density 1,000 kg/m^3 and oil of 25 mm and specific gravity 0.95 is placed on the water as shown in Figure Q1.

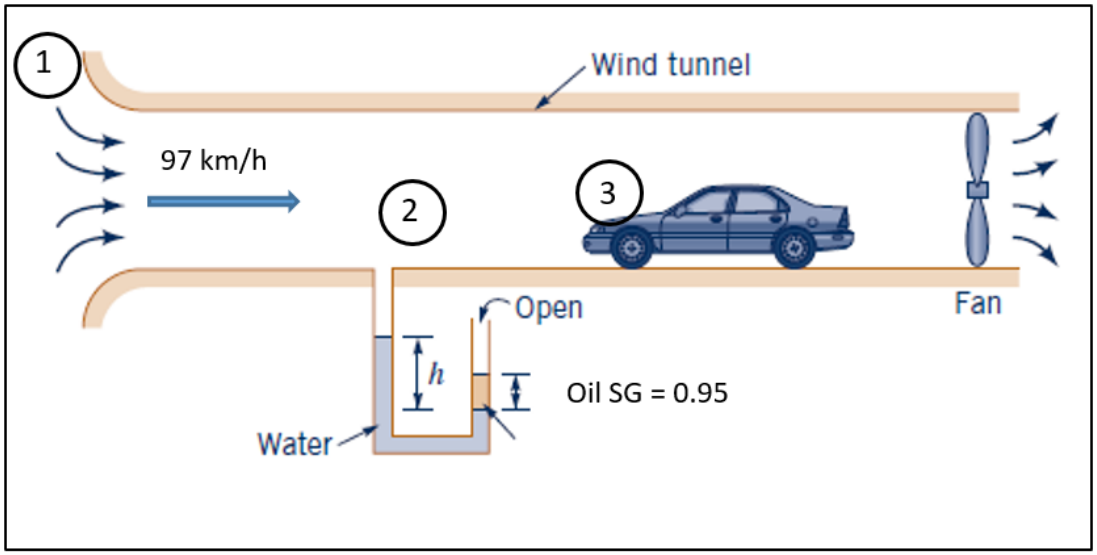

Figure Q1

With the velocity in the test section at 97 km/h, apply the Bernoulli Equation to answer Question 1a to Question 1e.

Question 1a

Interpret the static pressure (in Pascal) at the wind tunnel wall in the test section (Point 2).

(5 marks)

Question 1b

Determine the manometer reading h (m_{h2O}) (hint: the oil height is to be considered).

(5 marks)

Question 1c

Recognise the difference between the stagnation pressure on the front of the car (Point 3) and the static pressure in the test section (Point 2) in Pascal.

(7 marks)

Question 1d

When studying the flow around the car at high Reynold's Number, the stagnation pressure on the front of the car (Point 3) is found to be 1500 Pa and the static pressure in the test section (Point 2) is 500 Pa. Interpret the air velocity (km/h) in the test section (Point 2).

(5 marks)

Question 1e

Identify **THREE (3)** assumptions made in your results for Question 1.

The air density throughout the test can be taken as ρ_{air} = 1.225 kg/m^3 and the acceleration due to gravity is g = 9.81 m/s^2.

NOTE: All pressure measurements are in Gauge pressure.

(3 marks)

Question 2

A horizontal jet of water from a nozzle is deflected through an angle of 60° from its original direction by a stationary curved vane which it enters tangentially. The mass flow rate of water discharge from the nozzle is 1 kg/s. After calculation, it was found that the resultant force acting on the vane was 37.8 N at an angle of 53.4^o to the original direction of the jet, as shown in Figure Q2.

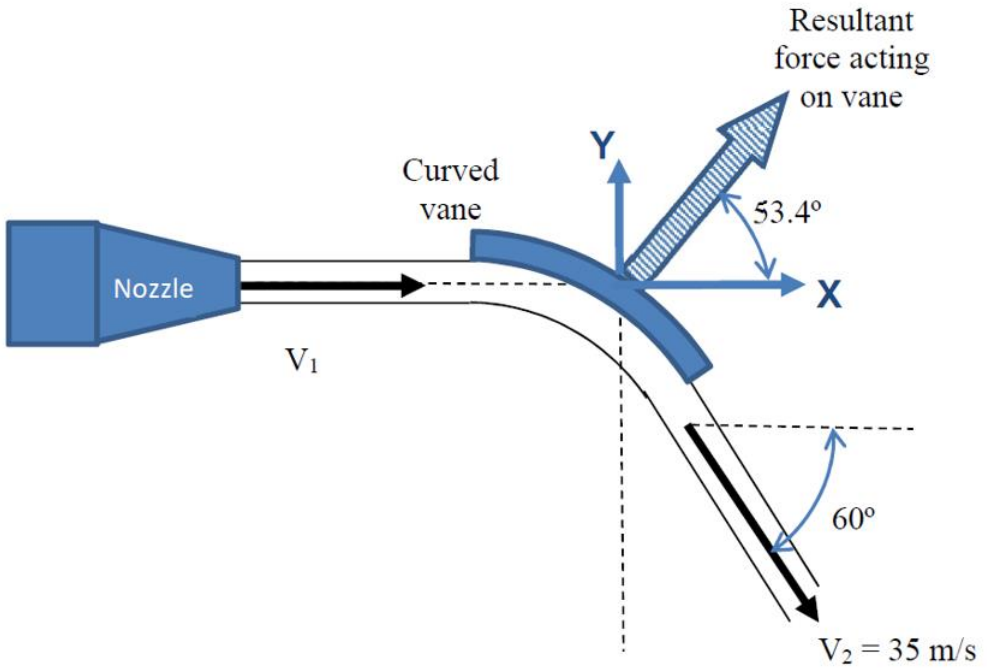

Figure Q2

Question 2a

Apply Newton's 2nd law of motion, recognise that the expression of force on water as:

$$F = \dot{m}(V_2 - V_1)$$

Identify the variables you use and explain **THREE (3)** assumptions in your derivation.

(8 marks)

Question 2b

Applying appropriate control volume, interpret the force (in N) in the X and Y directions acting on the fluid, based on the X and Y direction shown in Figure Q2. Indicate whether the forces are positive or negative.

(8 marks)

Question 2c

If the velocity V_2 leaving the vane is 35 m/s, interpret the velocity V_1 entering the vane.

(6 marks)

Question 2d

Evaluate the diameter of the nozzle exit (in mm).

(3 marks)

Question 3

A cargo plane with total mass of 80,000 kg has air-foil planform area of 300 m^2. The plane cruises at 378 km/h at an altitude of 12,000 m above sea level with air density at 0.312 kg/m^3. The cruising drag coefficient C_D of the plane is 0.032 while the maximum lift coefficient $C_{L,max}$ with and without flaps are 1.6 and 3.3 respectively.

The plane has double-slotted flaps for use during take-off and landing, but it cruises with all flaps retracted. The air density at sea level is taken 1.225 kg/m^3 and the take-off speed is 20% over the stall speed (safety factor of 1.2).

Question 3a

Present a derivation to recognise that the minimum velocity for take-off and landing is expressed as:

$$V_{min} = \sqrt{\frac{2W}{\rho C_{L,max} A}}$$

(4 marks)

Question 3b

Apply the expression you derived in Question 3a, evaluate the minimum safe velocity for take-off and landing <u>with</u> flaps and <u>without</u> the flaps.

(6 marks)

Question 3c

Using the Coefficient of Lift (C_L) vs Angle of Attack (α) distribution in Figure Q3c, identify the angle of attack (α) at cruising altitude.

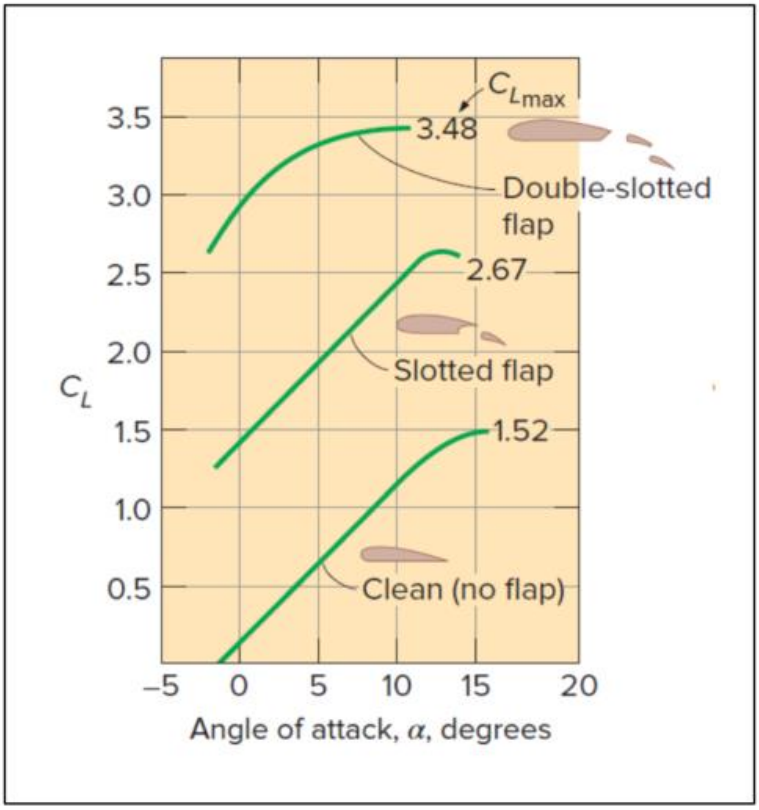

Figure Q3c

(5 marks)

Question 3d

Interpret the thrust (in kW) developed by the engine to overcome the wing drag.

(6 marks)

Question 3e

Briefly explain **TWO (2)** assumptions made in your calculations for Question 3.

(4 marks)

Question 4

Air enters an industrial gas-turbine power plant's compressor, operating on the Joule-Brayton Cycle, at atmospheric ambient conditions (28°C and 101 kN/m^2 absolute). The air is then compressed adiabatically in the compressor to a pressure of 808 kN/m^2 before entering the combustion chamber. The maximum cycle temperature reached is 1280°C at the outlet of the combustion chamber. The air then enters the turbine where it is expanded back to atmospheric conditions.

Question 4a

With appropriate sketches, using the Pressure vs Specific volume (p-v) and Temperature vs Entropy (T-s) relationships for this cycle, identify and describe the key components and processes taking place during the cycle.

(8 marks)

Question 4b

Apply suitable equations from the course to interpret the ideal temperatures, in K, at the end of the compression and expansion processes.

(4 marks)

Question 4c

Evaluate the amount of heat supplied and heat rejected in kJ/kg and using the results to interpret the thermal efficiency.

(8 marks)

Question 4d

Interpret the power output of the turbine unit in kW and the rate of heat input of this gas turbine in MW if the air flow rate is 2 kg/s.

Take Cp_{air} = 1.005 kJ/kgK and γ_{air}=1.4
Cp_{gas} = 1.15 kJ/kgK, and γ_{gas} = 1.33

(5 marks)

----- END OF PAPER -----